A Steganographic Covert Channel in the BB84 Classical Sifting Phase

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***Abstract*—This paper presents a steganographic covert channel embedded within the unencrypted, authenticated classical channel during the BB84 [1] sifting phase. Rather than altering the physical preparation of quantum states, risking the protocol's Information-Theoretic Security (ITS), this channel utilizes "classical misreporting." The sender maintains perfectly random quantum state preparation but intentionally falsifies specific basis announcements during sifting to transmit a covert payload. This method hides the covert data within the hardware's expected Quantum Bit Error Rate (QBER) noise floor, controlled by a rolling trigger sequence, rendering it statistically indistinguishable from natural errors within the limited sample size of a standard session. The channel inflates the observable QBER by a factor that halves with each increment of the trigger length k, remaining well below the protocol's abort threshold. The channel was empirically evaluated across trigger lengths $k = 6$ through $k = 10$ using Kolmogorov-Smirnov testing on both QBER distributions and inter-error distance distributions. At $k \geq 10$ the channel achieves statistical indistinguishability under both detection methods within 1,000 sessions in our experimental environment.**

I. Introduction

The benefits and security guarantees of Quantum Key Distribution (QKD) have been widely researched [1], [2], [3]. The protocols offer information-theoretic security derived from the properties of quantum physics. But while the quantum component of these protocols has been exhaustively analyzed, the classical channel has received less scrutiny as a potential carrier of hidden information.

The concept of subliminal channels in cryptographic protocols dates back to Simmons [4], who demonstrated that digital signature schemes could be exploited to carry hidden messages undetectable to outside observers. This work extends that tradition to the quantum domain: we explore whether the public and authenticated classical channel in QKD can be similarly exploited as a steganographic covert channel without disturbing the protocol's information-theoretic security.

Such a channel is relevant in scenarios where a compromised or coerced node in a QKD network seeks to exfiltrate small amounts of data without triggering alarms. In environments where an auditor (Eve) strictly monitors the network and whitelists traffic to exclusively permit QKD operations, standard encrypted communications (e.g., Signal) would be immediately flagged as anomalous. Unlike direct key leakage, which would be detectable through key verification failures or usage anomalies, a covert channel embedded in the sifting phase produces no observable effect beyond a minor QBER increase indistinguishable from hardware noise. This makes it a plausible vector for insider threats in multi-node QKD deployments.

We review BB84′s core design, propose our covert channel, discuss our implementation, and present empirical results from statistical analysis of the channel's detectability.

II. Background

In BB84 [1], Alice and Bob establish a shared secret key by exchanging qubits prepared in randomly selected conjugate bases. During the sifting phase, they publicly compare their basis choices over an authenticated classical channel, retaining only the bits where their bases matched. A subset of the sifted key is then sacrificed to estimate the Quantum Bit Error Rate (QBER), the fraction of compared bits that disagree. If the QBER exceeds approximately 11%, secure key extraction is no longer possible and the session is aborted [2]. While the quantum channel provides BB84′s information-theoretic security guarantees, the classical channel used for sifting is assumed to be public and authenticated but otherwise unprotected. It is this classical channel that this work targets for steganographic exploitation.

III. Design

The covert channel operates entirely at the classical software layer, leaving the physical quantum hardware and state preparation untouched. This section describes its core components.

A. Cryptographic Isolation

A dedicated, out-of-band Pre-Shared Key (PSK) is utilized for the covert channel. This PSK seeds two Pseudo-Random Number Generators (PRNG) synchronized between the sender (Alice) and receiver (Bob), one for the trigger sequences (PRNG_t) and one for the keystream (PRNG_k). Seeding them independently prevents the trigger sequence from being derivable from the keystream, and ensures the two PRNGs remain synchronized regardless of message length. In our implementation we use Python's `random.Random` (Mersenne Twister); however, in an actual implementation a cryptographically secure PRNG would be used.

B. Rolling Trigger Mechanism

The location of the covert bits is dictated probabilistically by PRNG_t to prevent statistical pattern analysis by Eve. PRNG_t generates a target sequence of k bases (e.g., for $k =$

7, a sequence such as $++\times+\times\times+$). Both Alice and Bob maintain a rolling buffer of the sifting announcements. The appearance of this trigger sequence indicates that the following basis announcement $(n+1)$ carries the covert payload. Immediately following a successful injection/extraction, both state machines query PRNG_t to generate a new trigger, ensuring a minimum spacing of $k+1$ between injections.

C. Injection Protocol

Alice executes standard BB84 state preparation, selecting her bases ($+$ or $\times$) and states (0 or 1) completely at random, preserving the core security of BB84.

Prior to the sifting phase, Alice prepares her message by attaching a preamble containing the message length as a fixed-width binary integer, with the width derived from the theoretical maximum channel capacity. The theoretical maximum capacity is $s/2^k$ covert bits, where s is the total number of qubits transmitted. The preamble width is the number of bits needed to represent this value. During the public sifting phase, Alice monitors her basis history. When her history matches the active trigger sequence, she replaces the basis announcement for qubit $n+1$.

Instead of announcing her true physical basis for qubit $n+1$, Alice announces the basis that corresponds to her covert ciphertext bit XOR'd with the next bit from PRNG_k, regardless of the actual qubit transmitted. She then updates her state machine to the next trigger. After the ciphertext is exhausted, she continues injecting random keystream bits from PRNG_k to maintain a constant injection rate regardless of message length.

D. Extraction Protocol

Bob extracts the covert message from Alice's basis announcements on the classical channel, independently of the sifting process. As Alice announces each basis, Bob feeds it into his synchronized state machine. Upon detecting the trigger sequence, Bob records Alice's next announced basis as a covert ciphertext bit, XOR'ing it with the next bit from PRNG_k. He then updates his state machine to the next trigger. This extraction operates on the complete announcement stream; bits that are later discarded during sifting still contribute to the covert channel.

After extraction is complete, he calculates the preamble width as described in Section III.C, reads the preamble to determine the message length, and extracts only the bits between the preamble and the end of the message (calculated from preamble length summed with the message length), discarding the remaining padding.

E. QBER Impact

Because Alice intentionally misreports her basis, she induces additional error into the primary quantum key. However, this artificial error should remain well below the system's standard 11% abort threshold [2], blending into standard hardware noise, although these numbers vary widely in practice [5].

Each covert injection follows a chain of three independent coin flips. First, because Alice deterministically replaces her basis announcement with a value mapped from the pseudo-random ciphertext, there is a 50% probability that this new value differs from the actual physical basis she used to prepare the photon. Second, given a genuine misreport, there is a 50% chance that Bob measured in a different basis than Alice announced; in this case the bit is discarded during sifting and again no error enters the key. Third, given a genuine misreport that survives sifting, Bob's measurement outcome is entirely uncorrelated with Alice's transmitted state, producing a bit error 50% of the time. The net error probability per injection is therefore $0.5\times0.5\times0.5=0.125$, or 12.5%. The total artificial QBER inflation (E_{covert}) is diluted by the rarity of the k-length trigger sequence occurring naturally, defined by probability $P=\left(\frac{1}{2}\right)^k$, giving

$$\mathrm{E}_{\mathrm{covert}} = 0.125 \times \left(\frac{1}{2}\right)^k. \tag{1}$$

Practical Hardware Impact: For a trigger length of $k=7$, the injection occurs once every 128 bits ($\approx 0.78\%$). The resulting artificial QBER inflation is:

$$E_{\mathrm{covert}} = 0.125 \times \left(\frac{1}{128}\right) \approx 0.098\%. \tag{2}$$

If the quantum hardware exhibits a standard noise floor of around 1.00% to 3.00% [6], the covert channel raises the total observable QBER to around 3.098%.

IV. Adversary Model

A. Adversary Capabilities

We assume a standard Dolev-Yao adversary model combined with full quantum interception capabilities. The adversary (Eve) has complete, passive read-access to the unencrypted classical channel (including Error Correction), the ability to intercept/measure/resend qubits, and full knowledge of the covert channel's algorithmic design. All four attack vectors described below are passive. An active adversary capable of delaying or replaying classical messages could potentially de-synchronize the PRNG state machines, causing Alice and Bob to diverge on trigger positions and producing extraction failures detectable as anomalous message corruption. Resistance to active classical-channel attacks is outside the scope of this work and remains an open problem.

B. Attack Vector 1: Global Basis Distribution

Eve searches for an imbalance in the ratio of Rectilinear ($+$) to Diagonal ($\times$) announcements.

Defense: The covert message is XOR'd with a PRNG keystream prior to transmission, whitening the ciphertext into a uniform pseudo-random distribution. The global distribution of announced bases remains exactly 50/50.

C. Attack Vector 2: Error-Correlation Analysis

Eve monitors the public Error Correction phase to pinpoint exact error indices, mapping them back to the sifting phase to isolate the trigger sequence preceding artificial errors (E_{covert}).

Defense: Because the PRNG synchronously advances the state machine after every injection, the active trigger sequence

is drawn uniformly at random from the space of all possible 2^k sequences. To Eve's statistical analysis tools, the preceding basis history H for an artificial error ($e_{\text{covert}} \in E_{\text{covert}}$) is mathematically indistinguishable from the preceding basis history for a natural hardware error ($e_{\text{natural}} \in E_{\text{natural}}$).

Let S represent any specific, arbitrary sequence of k bases chosen by the adversary for statistical testing. Because the active trigger is drawn from a uniform pseudo-random distribution, the probability of sequence S preceding a covert error is identical to the probability of it preceding a natural hardware error:

$$P(H(e_{\text{covert}}) = S) = \frac{1}{2^k} = P(H(e_{\text{natural}}) = S). \quad (3)$$

Eve cannot computationally differentiate the PRNG output from standard quantum noise within the limited sample size of a standard session, rendering the channel effectively integrated into the baseline noise of the physical hardware.

D. Attack Vector 3: QBER Distribution Analysis

An adversary with access to QBER measurements from multiple QKD sessions could compare the observed QBER distribution against an expected baseline for the hardware. If the covert channel consistently inflates the QBER, even by a small amount, this shift can become statistically detectable given enough observations. We evaluate this attack using the two-sample Kolmogorov-Smirnov (KS) test, which measures the maximum distance between the empirical cumulative distribution functions of two samples. The KS test is well-suited to this analysis because it makes no assumptions about the underlying distribution shape. We apply it with a significance level of $\alpha = 0.05$, where a p-value below this threshold indicates the two distributions are distinguishable. Results across varying trigger lengths and session counts are presented in Section VI.

E. Attack Vector 4: Inter-Error Distance Analysis

A more sophisticated adversary could analyze the positions of errors within the sifted key rather than just the overall error rate. Because the trigger mechanism imposes a minimum spacing of $k + 1$ between covert injections, the resulting error pattern may differ structurally from purely random hardware errors. However, our evaluation indicates that the KS test's sensitivity is not primarily driven by this structural spacing, but rather by the reduction in the mean inter-error distance. Because the covert channel introduces a higher absolute volume of errors, the average distance between them inherently shrinks. The KS test easily flags this shift because of the massive statistical power afforded by analyzing thousands of inter-error distances per session, rather than a single aggregate QBER metric.

V. Implementation

The covert channel was implemented as an extension of an existing BB84 simulation by Bhatnagar [7], which provides object-oriented implementations of Alice, Bob, and the quantum and classical channels. The simulation models qubit preparation, transmission, measurement, basis sifting, and key validation, with support for channel noise and eavesdropping scenarios.

Rather than modifying the base classes directly, we followed the existing codebase's pattern of using inheritance to extend functionality. A `CovertStateMachine` class encapsulates the rolling trigger mechanism, maintaining the PRNG instances, the trigger buffer, and the keystream generator. Both the covert Alice and covert Bob subclasses instantiate their own `CovertStateMachine` with the same PSK and trigger length, ensuring synchronized operation.

The covert Alice subclass adds a `misreport` method that iterates through the basis sequence after quantum state preparation but before the sifting announcement, replacing bases at trigger positions with covert ciphertext bits. The covert Bob subclass adds an `extract_msg` method that scans the received basis announcements for trigger matches and collects the subsequent bits. Neither subclass modifies any quantum-layer behavior — state preparation, transmission, and measurement remain identical to the base implementation.

For the noise model, we subclassed the existing quantum channel to accept a configurable error rate rather than using the original hardcoded value. The noise model applies random X errors (bit flips) to a fraction of transmitted qubits, simulating depolarizing channel noise. The Hadamard gate failure model present in the original codebase was not used, as our analysis focuses on channel noise as the primary source of baseline QBER.

The experimental driver comprises these components: covert Alice and Bob with the noisy quantum channel, running the standard BB84 phases in sequence with the misreporting and extraction steps inserted between quantum transmission and basis announcement. The complete implementation is available at [8].

VI. Results

All experiments were conducted with a qubit stream length of $s = 65,536$, a channel error rate of 3%, and 1000 independent trials per trigger length. Baseline experiments used identical parameters without the covert channel active. The results are summarized in the table below.

TABLE I

SUMMARY OF EXPERIMENTAL RESULTS ACROSS TRIGGER LENGTHS $k = 6$ THROUGH $k = 10$.[1]

k	Baseline QBER	Covert QBER	Gap	QBER p-value	Inter-Error p-value	QBER Det. @
6	0.7519%	0.9217%	0.1697%	8.03×10^{-96}	$< 10^{-10}$	20
7	0.7405%	0.8462%	0.1056%	1.01×10^{-37}	$< 10^{-10}$	30
8	0.7480%	0.8005%	0.0524%	1.71×10^{-10}	$< 10^{-10}$	50
9	0.7571%	0.7725%	0.0154%	0.078	0.001	>1000
10	0.7567%	0.7652%	0.0085%	0.148	0.422	>1000

A. QBER Validation

The observed QBER inflation closely follows the corrected theoretical prediction of $E_{\text{covert}} = 0.125 \times \left(\frac{1}{2}\right)^k$. As shown in Fig. 1, the measured gap between baseline and covert QBER roughly halves with each increment of k, declining from 0.1697% at $k = 6$ to 0.0085% at $k = 10$. The observed values fall slightly below the theoretical prediction, which we attribute to the feedback effect of misreporting on the trigger mechanism: modified basis values feed back into the rolling buffer, occasionally disrupting subsequent trigger matches and reducing the effective injection rate below the theoretical maximum of $s/2^k$ as seen in Table II, with standard deviation decreasing with increased k values.

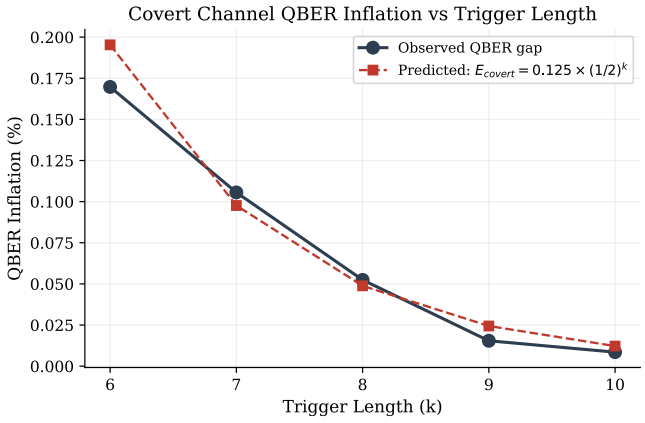


Fig. 1. Observed QBER inflation compared to the theoretical prediction across trigger lengths $k = 6$ through $k = 10$.

[1]QBER Det. @ indicates the minimum number of observed sessions at which the QBER-based KS test first reaches significance at $\alpha = 0.05$.

TABLE II

COVERT CHANNEL CAPACITY: THEORETICAL MAXIMUM VERSUS OBSERVED MEAN.[2]

k	Theoretical Capacity (bits)	Observed Capacity (bits)	Yield	Std. Dev.
6	1024.0	951.0	92.9%	28.9
7	512.0	489.6	95.6%	20.5
8	256.0	251.0	98.0%	15.2
9	128.0	125.7	98.2%	10.3
10	64.0	64.2	100.3%	7.9

B. QBER Detectability

We applied the two-sample KS test to the baseline and covert QBER distributions at increasing session counts to determine how many observations an adversary would need to detect the channel. As shown in Fig. 3 (Appendix A), the detectability threshold varies significantly with k. At $k = 6$ the channel is detectable in under 20 sessions, $k = 7$ at around 30 sessions, and $k = 8$ at 50 sessions. At $k = 9$ and $k = 10$, the QBER distributions remain statistically indistinguishable at all tested session counts, with a p-value of 0.078 and 0.148, respectively, at 1000 sessions. The QBER histograms in Fig. 4 (Appendix A) illustrate this visually: at $k = 6$, the baseline and covert distributions are clearly separated, while by $k = 9$ they overlap almost entirely.

C. Inter-Error Distance Analysis

While QBER analysis becomes ineffective at $k \geq 9$, the inter-error distance analysis proves to be a more sensitive detection method. By computing the distances between consecutive errors in the sifted key and comparing them across baseline and covert experiments, we find that the covert channel remains detectable at trigger lengths up to $k = 9$, as shown in Fig. 2. At $k = 10$, both analyses fail to detect the channel — QBER analysis yields $p = 0.148$ and inter-error distance analysis yields $p = 0.422$, indicating that $k = 10$ represents the threshold at which the covert channel achieves full statistical indistinguishability under both methods within our experimental scope.

This increased sensitivity of the inter-error distance analysis is driven primarily by the difference in total error count rather than a structural spacing pattern from the trigger mechanism. The covert channel introduces additional errors, reducing the mean inter-error distance. At $k = 6$, the mean baseline distance is 131.8 bits compared to 107.7 bits for the covert channel. This gap narrows with increasing k values, at $k = 10$, the means are 130.8 and 129.6, respectively, and the KS test can no longer distinguish them.

[2]Yield represents the fraction of theoretical capacity achieved.

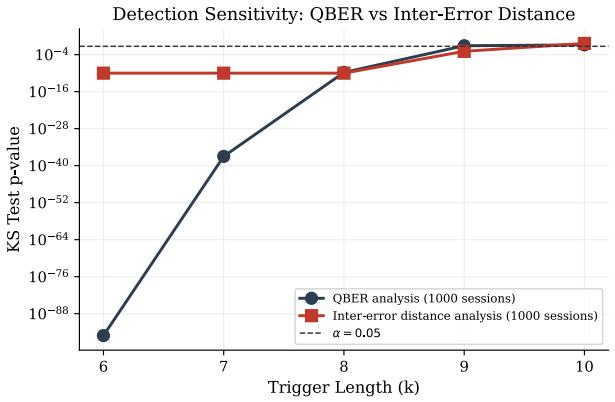


Fig. 2. Comparison of KS test p-values for QBER analysis versus inter-error distance analysis at 1000 sessions.

VII. Discussion

Our evaluation demonstrates that the proposed covert channel is stealthier than initially predicted. Because half of all misreported announcements coincidentally match the physical preparation basis, the effective QBER inflation is heavily suppressed. This creates a highly favorable tradeoff between bandwidth and detectability.

The detectability analysis reveals two distinct attack axes with different sensitivity thresholds. QBER-based detection becomes infeasible at $k \geq 9$ within our 1,000-session scope, but inter-error distance analysis remains effective at all but one of the tested k values. Practically, this distinction matters; an adversary monitoring aggregate QBER per session faces a significantly harder detection problem than one with access to individual error positions from the error correction phase. The central tradeoff is ultimately bandwidth versus stealth: at $k = 6$, approximately 951 covert bits per session are available but trivially detectable, whereas at $k = 10$, bandwidth drops to approximately 64 bits but both QBER and inter-error detection are neutralized.

This work has several limitations that define the boundaries of our current model. Theoretically, the simulation does not utilize a cryptographically secure PRNG, and the inter-error distance analysis assumes the adversary has complete read-access to individual error positions—a capability that may not hold across all reconciliation protocols. Physically, the simulation relies on a uniform distribution of random X errors, failing to capture real-world hardware effects like dark counts, phase drift, or detector dead-time. Dead-time is particularly relevant; because real QKD systems naturally suppress closely clustered errors, an adversary might detect the rigid $k + 1$ minimum spacing of our rolling trigger by observing clashes with the hardware's natural error clustering. Future work should validate the channel against real hardware noise profiles and explore an adaptive trigger mechanism—where Alice dynamically scales k in response to real-time error density—alongside formal indistinguishability proofs and testing against machine-learning-based detection.

VIII. Applicability to Other QKD Protocols

The covert channel described in this work exploits three structural properties of BB84′s classical post-processing: discrete basis announcements, a QBER tolerance sufficient to absorb the covert inflation, and independent processing of the announcement stream by both parties. Any QKD protocol exhibiting all three properties is a candidate for the same technique.

The six-state protocol [9] and decoy-state BB84 [10] both retain BB84′s sifting structure and are directly applicable. SARG04 [11], MDI-QKD [12], and twin-field variants [13] use modified announcement structures that would require adapting the injection mechanism but do not fundamentally prevent the approach. Entanglement-based protocols like E91 [14] do announce discrete bases during sifting, but the symmetric structure where both parties independently choose measurement bases on entangled pairs changes the error-propagation model and would require adapting the injection mechanism. CV-QKD [15] lacks discrete basis announcements entirely, making the technique inapplicable. Recent sifting-free protocols eliminate the attack surface altogether.

The covert channel thus generalizes readily to prepare-and-measure protocols with discrete basis sifting, which remain the most widely deployed family of QKD protocols.

IX. Conclusion

We presented a steganographic covert channel embedded in the BB84 classical sifting phase that operates entirely at the software layer without disturbing the quantum channel. We derived and empirically validated a corrected QBER inflation formula, and characterized a two-axis detection landscape in which QBER analysis and inter-error distance analysis offer different sensitivity thresholds. The trigger length k provides a tunable tradeoff between covert bandwidth and resistance to detection. These findings demonstrate that the classical post-processing phases of QKD protocols represent an under-examined attack surface, and that security analyses focused exclusively on the quantum channel may overlook exploitable information leakage at the classical layer.

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AI tools were used as a writing assistant during manuscript preparation. All research design, experimental analysis, and scientific conclusions are the work of the author.

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Appendix A: Additional Figures

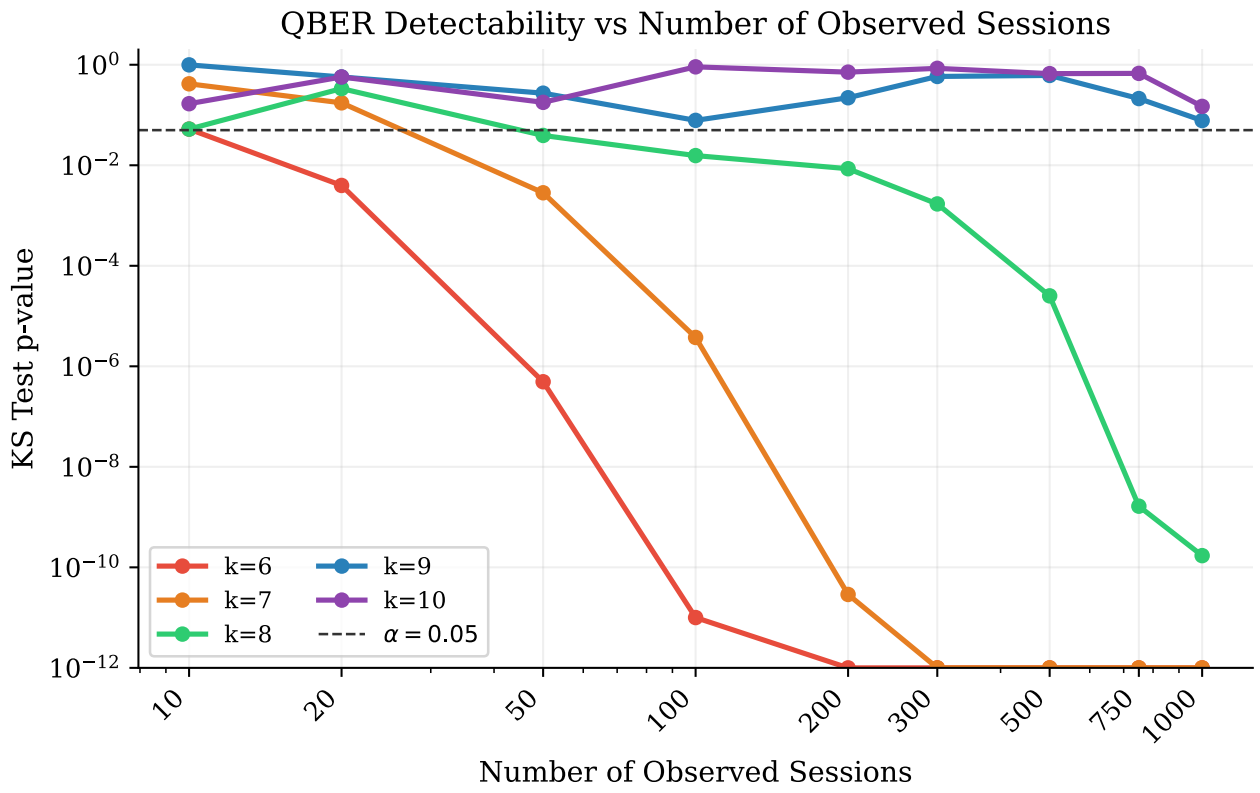


Fig. 3. KS test p-value versus number of observed sessions for each trigger length. The dashed line indicates the $\alpha = 0.05$ significance threshold.

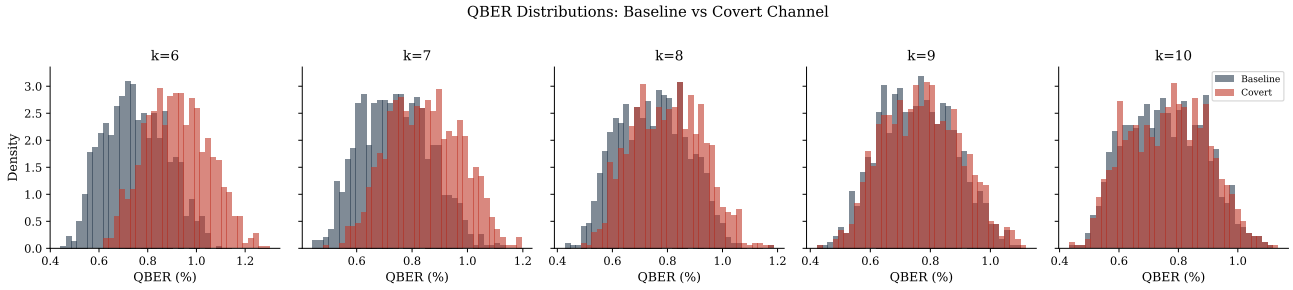


Fig. 4. QBER distributions for baseline (blue) and covert (red) experiments across trigger lengths $k = 6$ through $k = 10$.

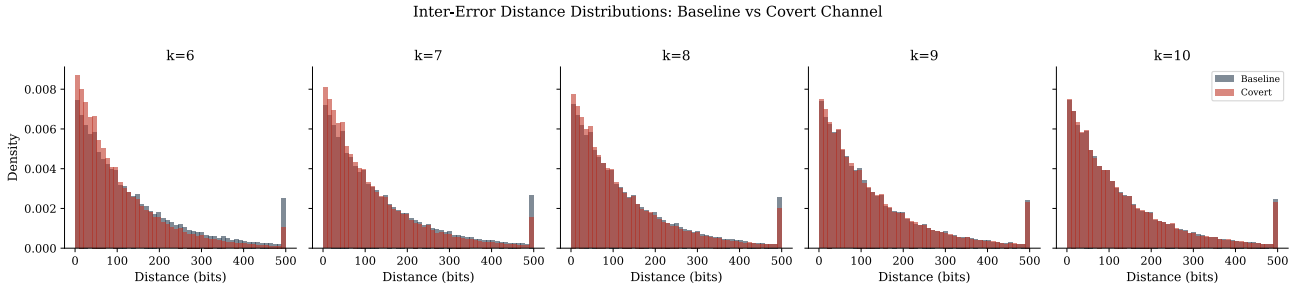


Fig. 5. Inter-error distance distributions for baseline and covert experiments across trigger lengths.

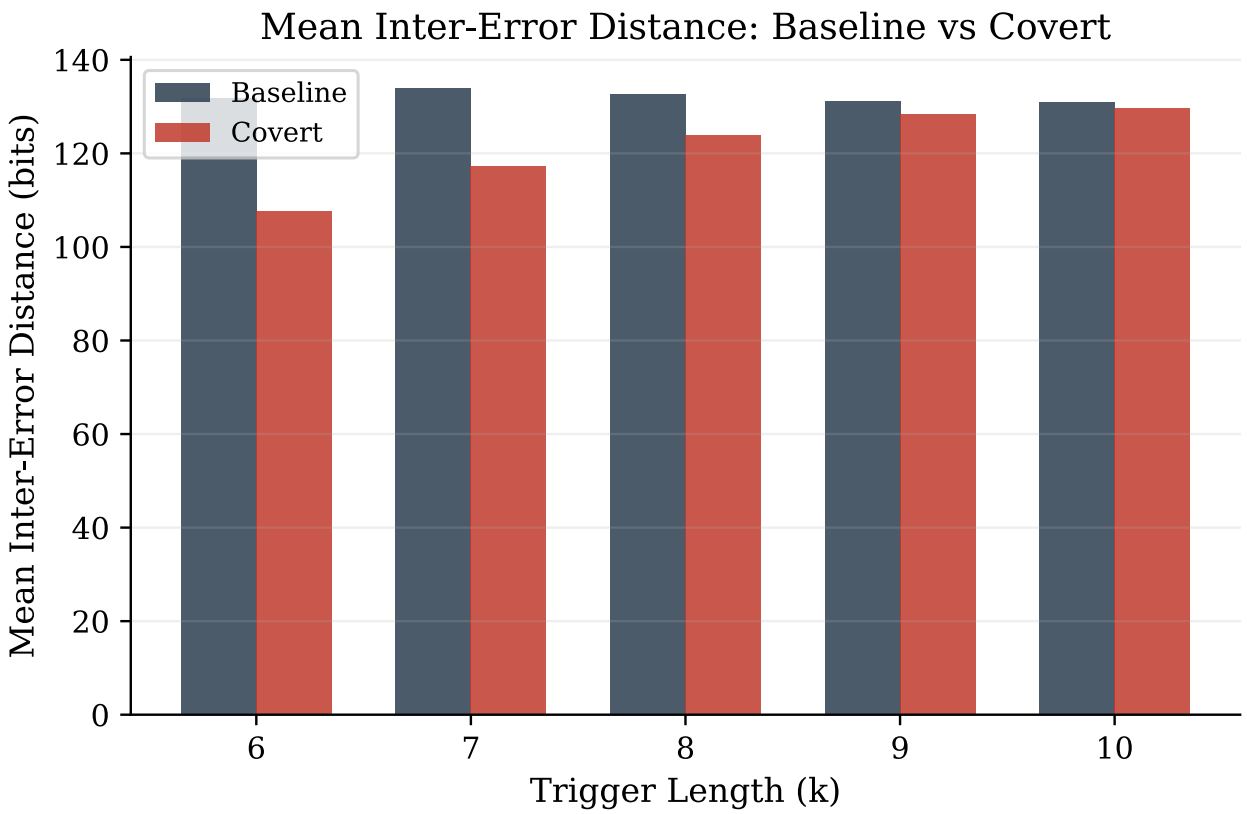


Fig. 6. Mean inter-error distance for baseline and covert experiments across trigger lengths.